Securing AI and Advanced Topics

Today's goal • 3/3 items

Module 1
Course Introduction

Module 2
Fraud Prevention with Cloud AI Solutions

Module 3
Introduction to Generative Adversarial Attacks (GANs)

Module 4
GANs and Adversarial Attacks

Understanding Black-box and White-box Adversarial Attacks

- Adversarial Attacks Explained
  Video • 8 min
- Reading References
  Reading • 10 min
- Understanding Black-box and White-box Adversarial Attacks
  Practice Assignment • Grade: 100%

Practical Implementation of Adversarial Attacks

- Hands-On Adversarial Attacks
  Video • 9 min
- Reading References
  Reading • 10 min
- Practical Implementation of Adversarial Attacks
  Practice Assignment • Grade: 100%

Module-end Assessments

- **Self-Reflective Reading: GANs and Adversarial Attacks**
  Reading • 45 min
- Practice Lab: Generating Synthetic QR Codes with the Trained Generator
  Lab • 1h
- GANs and Adversarial Attacks
  Graded Assignment • 1h

Module 5
Reinforcement Learning

Reinforcement Learning and Its Applications

- Reinforcement Learning
  Video • 8 min
- Reading References
  Reading • 10 min
- Reinforcement Learning and its Applications
  Practice Assignment • 15 min

Using RL for Adversarial Attacks and Optimizing Datasets

- Evading Malware Detectors with RL
  Video • 6 min
- Reading References
  Reading • 10 min
- Using RL for Adversarial Attacks and Optimizing Datasets
  Practice Assignment • 15 min

Module-end Assessments

- Self-Reflective Reading: Understanding Reinforcement Learning
  Reading • 45 min
- Reinforcement Learning
  Graded Assignment • 1h

Module 6
Evaluating AI Models and Performance

Feature Engineering and Model Optimization Techniques

- Challenges with Using AI for Cybersecurity
  Video • 8 min
- Reading References
  Reading • 10 min
- Feature Engineering and Model Optimization Techniques
  Practice Assignment • 15 min

Evaluating AI Models and Performance Metrics

- Evaluating AI Models
  Video • 7 min
- Reading References
  Reading • 10 min
- Evaluating AI Models and Performance Metrics
  Practice Assignment • 15 min

Module-end Assessments

- Self-Reflective Reading: Feature Engineering in Cybersecurity Applications
  Reading • 45 min
- Evaluating AI Models and Performance
  Graded Assignment • 1h

Self-Reflective Reading: GANs and Adversarial Attacks

0:06 / 3:12 2x

Introduction

In this module, "GANs and Adversarial Attacks," we delve into the intriguing world of adversarial examples and their implications for machine learning models. Adversarial examples are inputs to machine learning models that an attacker has intentionally designed to cause the model to make a mistake. This phenomenon raises significant concerns in various applications, particularly in the field of cybersecurity.

Task Overview

You are tasked with conducting a literature review on adversarial examples and methods, specifically focusing on the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD). Your review should highlight the differences between these methods and the latest techniques in adversarial example generation. Additionally, you will explore how advanced adversarial example methods can benefit cyber analysts and provide practical examples. Furthermore, you will summarize the development of adversarial examples utilizing Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs), using terminology from the ML analytic development process discussed in the lecture. You will also investigate the performance tuning parameters in the provided Jupyter notebook and attempt to optimize them to yield better results.

Suggested Steps

1. **Conduct a Literature Review**:
   - Research adversarial examples, focusing on FGSM and PGD.
   - Identify and summarize the differences between these methods and the latest advancements in adversarial techniques.
2. **Evaluate the Benefits for Cyber Analysts**:
   - Discuss how advanced methods for creating adversarial examples can assist cyber analysts.
   - Provide specific examples where applicable.
3. **Summarize Development Using DNNs and CNNs**:
   - Use the ML analytic development process language to outline the development of adversarial examples from your lecture.
4. **Investigate the Jupyter Notebook**:
   - Open the notebook located in chapter 8 (adversarial_examples/adversarial_examples.ipynb.)
   - Identify performance tuning parameters and describe the steps you take to optimize these parameters for improved results.

Self-Reflective Questions

1. **Literature Review**:
   - What key differences did you discover between FGSM, PGD, and more recent adversarial methods?
   - How do these differences impact the effectiveness and applicability of adversarial examples in real-world scenarios?
2. **Benefits for Cyber Analysts**:
   - In what ways can advanced adversarial methods enhance the capabilities of cyber analysts in threat detection?
   - Can you think of a situation where understanding these methods could directly influence a cybersecurity strategy?
3. **Development Summary**:
   - How did the language from the ML analytic development process help clarify the process of creating adversarial examples?
   - What insights did you gain about the relationship between DNNs/CNNs and adversarial attacks?
4. **Jupyter Notebook Investigation**:
   - What performance tuning parameters did you identify, and why are they significant for the optimization process?
   - What challenges did you encounter while attempting to optimize these parameters, and how did you overcome them?

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